Kenuo **Xu**

SENIOR ENGINEER · HUAWEI

Building A9b, Huawei Lianqiu Lake R&D Center, Shanghai, China 201718

(+86) 13269762233 | kenuo.xu@foxmail.com | https://witty-me.github.io/ | Witty-me | Google Scholar

Employment

Huawei *Shanghai, China*

SENIOR ENGINEER (WLAN LAB, DATA COMMUNICATION PRODUCT LINE) *2025.08 - Present*

- Led the design and development of end-to-end AI architectures for physical-world perception through Wi-Fi CSI-based sensing, spanning the pre-research, development, and deployment of intelligent sensing models for smart security and smart building applications.
- Developed lightweight CNN-based sensing models for edge deployment, improving robustness to data imbalance and environmental domain shifts through training-time data augmentation and ADDA-based domain adaptation techniques.
- Designed a Transformer Encoder-based self-supervised representation learning framework for Wi-Fi CSI, incorporating CSI-specific tokenization and physics-guided pretraining objectives to learn transferable representations across sensing environments and downstream tasks.
- Collaborated cross-functionally to enhance and integrate signal-processing methods into the AI sensing pipeline, optimize and deploy models on edge devices, and drive real-world validation and product integration.
- Received an "Excellent" rating upon completion of probation.

Microsoft Research Asia *Shanghai, China*

RESEARCH INTERN (SHANGHAI WIRELESS GROUP; MENTOR: PROF. LILI QIU) *Dec. 2022 - Sep. 2023*

- Conducted cross-disciplinary research at the intersection of Large Language Models (LLMs) and computer networking systems, exploring the potential of applying LLMs to underlying system optimizations.
- Designed and implemented an LLM-driven intelligent parameter-tuning framework for network algorithms, leveraging the model's comprehension and reasoning capabilities to replace manual tuning.

Education

Peking University *Beijing, China*

PH.D. IN COMPUTER SCIENCE *Sep. 2020 - Jun. 2025*

- In the Software-hardware Orchestrated ARchitecture (SOAR) group; advisor: Prof. Chenren Xu
- Research interest: algorithms of mobile computingn and their applications in IoT, robotics, and human-computer interaction.
- Designed a visible light backscatter communication system that supports concurrent transmission for low latency purpose.
- Designed a visible light communication system with spike cameras as receivers to achieve high data rate and dynamic range.
- Designed a liquid-crystal fiducial marker system using LiDAR as receivers for extended reading range and higher ranging accuracy.
- See publications for more research.

Peking University *Beijing, China*

B.SC. IN COMPUTER SCIENCE *Sep. 2016 - Jun. 2020*

- Graduated with Excellent Graduate Award.

Publications

RetroLiDAR: A Liquid-crystal Fiducial Marker System for High-fidelity Perception of Embodied AI *ACM SenSys*

KENUO XU, BO LIANG, JINGYU LI, CHENREN XU *2025*

- A long-range high-ranging-accuracy fiducial marker system using LiDAR and liquid crystals for robotics and virtual reality.

RetroV2X: A New V2X Paradigm with Visible Light Backscatter Networking *Fundamental Research*

CHENREN XU, **KENUO XU**, LILEI FENG, BO LIANG *2023*

- A practical vehicle-to-everything communication system with visible light.

When Visible Light (Backscatter) Communication Meets Neuromorphic Cameras in V2X *ACM HotMobile*

KENUO XU, KEXING ZHOU, CHENGXUAN ZHU, SHANGHANG ZHANG, BOXIN SHI, XIAOQIANG LI, TIEJUN HUANG, CHENREN XU *2023*

- When VLC meets neuromorphic cameras: a spike cameras as VLC receiver achieves 4.8 kbps data rate and different mobile scenarios.

Low-Latency Visible Light Backscatter Networking with RetroMUMIMO *ACM SenSys*

KENUO XU, CHEN GONG, BO LIANG, YUE WU, BOYA DI, LINGYANG SONG, CHENREN XU *2022*

- Enables 8 concurrent VLBC links and reduces networking latency by 92.0%.

LightRider: Reliable UAV Ground Communication with a Single Laser Tethering Link *IEEE SECON*

JUNJIE WANG, **KENUO XU**, ZHE OU, ZHAOFENG LUO, BO LIANG, MUHAN LI, XIAOMING LIU, LINGYANG SONG, GUOBIN SHEN, XINWEI YAO, CHENREN XU *2026*

- An self-laser-tethering system for free-space optical communition with high mobility.

EchoSight: Streamlining Bidirectional Virtual-physical Interaction with In-situ Optical Tethering *ACM CHI*

Jingyu Li, Qingwen Yang, **Kenuo Xu**, Yang Zhang, Chenren Xu *2025*

- An optical look-and-control user interaction system for AR glasses.

High-Speed Passive Visible Light Communication with Event Cameras and Digital Micro-Mirror *ACM SenSys*

Yanxiang Wang, Yiran Shen, **Kenuo Xu**, Mahbub Hassan, Guangrong Zhao, Chenren Xu, Wen Hu *2024*

- 16x throughput improvement for passive VLC using event camera and digital micro-mirror device.

Designing Network Algorithms via Large Language Models *ACM HotNets*

Zhiyuan He, Aashish Gottipati, Lili Qiu, Xufang Luo, **Kenuo Xu**, Yuqing Yang, Francis Y. Yan *2024*

- Using LLMs to design algorithms tailored for computer networks.

VLID: Visible Light Backscatter System for Battery-free Internet-of-Things *IEEE/ACM Transactions on Networking*

Chenren Xu, Purui Wang, Tuochao Chen, Yue Wu, **Kenuo Xu**, Xieyang Xu, Yang Shen, Junrui Yang, Guojun Chen, Guobin Shen *Accepted*

- An end-to-end VLBC solution for battery-free IoT networking.

Renovating road signs for infrastructure-to-vehicle networking: a visible light backscatter communication and networking approach *ACM MobiCom*

Purui Wang, Lilei Feng, Guojun Chen, Chenren Xu, Yue Wu, **Kenuo Xu**, Guobin Shen, Kuntai Du, Gang Huang, Xuanzhe Liu *2020*

- Enhance the reliability of autonomous driving with reflective roadsigns that conveys dynamic additional information.

Turboboosting Visible Light Backscatter Communication *ACM SIGCOMM*

Yue Wu, Purui Wang, **Kenuo Xu**, Lilei Feng, Chenren Xu *2020*

- Improve the data rate of VLBC by 8x (prototype) and 32x (simulation) with advanced modulation schemes.

Honors & Awards

2024	**Youth Award for Athletics**, Peking University	*Beijing, China*
2024	**Second Prize**, Ubiquitous Intelligent Sensing Technology Innovation and Application Competition	*Hangzhou, China*
2022	**Merit Student**, Peking University	*Beijing, China*
2021	**First Prize**, Competition of Future Network Technology Innovation	*Nanjing, China*
2020	**Excellent Graduate**, Peking University	*Beijing, China*
2019	**Houston BAA Scholarship**, Peking University	*Beijing, China*
2019	**Merit Student**, Peking University	*Beijing, China*

Activities

Teaching Assistant *Peking University*

Computer Networks (Honor Track) *Fall 2019, 2020, 2021-2024(Light)*

- Organizing the course and answering questions.
- Giving assignments, tutorials, and grading of labs.
- Designing quizzes and grading students' responses.
- Mentoring course research projects (light).

Journal Reviewer

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) *2021*